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10 / 10 submitted

Q1. Second Largest

Score: 10.00 TC: 3/3 passed

CODE

```
#include<stdio.h>
int main(){
    int a[5];
    int largest , second;

    for(int i=0;i<5;i++){
        scanf("%d",&a[i]);
    }
    largest = second = -999999;

    for(int i=0;i<5;i++){
        if(a[i] >largest){
            second = largest;
            largest = a[i];
        }
        else if (a[i]>second && a[i]≠largest){
            second = a[i];
        }
    }
    printf("%d",second);

    return 0;
}
```

INPUT

10 5 8 20 15

OUTPUT

15

Q2. Missing Number

Score: 10.00

TC: 3/3 passed

CODE

```
#include<stdio.h>
int main(){
    int a[4];
    int sum = 0;
    int total = 15;

    for(int i=0;i<4;i++){
        scanf("%d",&a[i]);
        sum = sum + a[i];
    }
    printf("%d",total - sum);

    return 0;
}
```

INPUT

1 2 4 5

OUTPUT

3

Q3. Longest Unique Substring

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[1000];
    scanf("%99s", s);

    int n = strlen(s);
    int maxLen = 0;

    for (int i = 0; i < n; i++) {
        int visited[256];
        for (int k = 0; k < 256; k++) {
            visited[k] = 0;
        }

        int len = 0;
        for (int j = i; j < n; j++) {
            int ch = s[j];
            if (visited[ch] == 1) {
                break;
            }
            visited[ch] = 1;
            len = len + 1;
        }

        if (len > maxLen) {
            maxLen = len;
        }
    }

    printf("%d\n", maxLen);
    return 0;
}
```

INPUT

abcabcbb

OUTPUT

3

Q4. Majority Element

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int a[1000], n = 0;
    int i, candidate, count = 0, frequency = 0;

    while (scanf("%d", &a[n]) == 1)
        n++;

    if (n == 0) {
        printf("No Majority Element");
        return 0;
    }

    /* Find candidate */
    for (i = 0; i < n; i++) {
        if (count == 0) {
            candidate = a[i];
            count = 1;
        }
        else if (a[i] == candidate) {
            count++;
        }
        else {
            count--;
        }
    }

    /* Verify candidate */
    for (i = 0; i < n; i++) {
        if (a[i] == candidate)
            frequency++;
    }

    if (frequency > n / 2)
        printf("%d", candidate);
    else
        printf("No Majority Element");

    return 0;
}
```

INPUT

3 3 4 2 3 3 3

OUTPUT 3

Q5. Palindrome String

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>
#include <string.h>

int main() {
    char str[1000];
    int i, len;
    int isPalindrome = 1;

    scanf("%99s", str);

    len = strlen(str);

    for (i = 0; i < len / 2; i++) {
        if (str[i] != str[len - i - 1]) {
            isPalindrome = 0;
            break;
        }
    }

    if (isPalindrome) {
        printf("Palindrome");
    } else {
        printf("Not palindrome");
    }

    return 0;
}
```

INPUT

madam

OUTPUT

Palindrome

Q6. Sort 0 1 2

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    int a[1000];
    int n = 0;
    int i, j, temp;

    while (scanf("%d", &a[n]) == 1)
    {
        n++;
    }

    for (i = 0; i < n - 1; i++)
    {
        for (j = i + 1; j < n; j++)
        {
            if (a[i] > a[j])
            {
                temp = a[i];
                a[i] = a[j];
                a[j] = temp;
            }
        }
    }

    for (i = 0; i < n; i++)
    {
        if (i > 0)
            printf(" ");

        printf("%d", a[i]);
    }

    return 0;
}
```

INPUT

2 0 2 1 1 0

OUTPUT

0 0 1 1 2 2

Q7. Longest Palindrome

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    char str[1000];
    int n = 0;
    int i, left, right;
    int start = 0, maxLen = 1;
    int len;

    scanf("%99s", str);

    while (str[n] != '\0')
    {
        n++;
    }

    for (i = 0; i < n; i++)
    {
        left = i;
        right = i;

        while (left ≥ 0 && right < n && str[left] == str[right])
        {
            len = right - left + 1;

            if (len > maxLen)
            {
                maxLen = len;
                start = left;
            }

            left--;
            right++;
        }

        left = i;
        right = i + 1;

        while (left ≥ 0 && right < n && str[left] == str[right])
        {
            len = right - left + 1;
```

```
        if (len > maxLen)
        {
            maxLen = len;
            start = left;
        }

        left--;
        right++;
    }
}

for (i = start; i < start + maxLen; i++)
{
    printf("%c", str[i]);
}

return 0;
}
```

INPUT

babad

OUTPUT

bab

Q8. Maximum Subarray

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main() {
    int a[1000], n = 0;
    int i, current, maximum;

    while (scanf("%d", &a[n]) == 1) {
        n++;
    }

    if (n == 0)
        return 0;

    current = a[0];
    maximum = a[0];

    for (i = 1; i < n; i++) {
        if (current + a[i] > a[i])
            current = current + a[i];
        else
            current = a[i];

        if (current > maximum)
            maximum = current;
    }

    printf("%d", maximum);

    return 0;
}
```

INPUT

-5 -2 -8 -1

OUTPUT

-1

Q9. Search in Rotated Sorted Array

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    int a[1000], n = 0;
    int target, i;
    char ch;

    while (scanf("%d", &a[n]) == 1)
    {
        n++;

        ch = getchar();

        if (ch == ';')
            break;

        if (ch == EOF)
            break;

        ungetc(ch, stdin);
    }

    scanf("%d", &target);

    for (i = 0; i < n; i++)
    {
        if (a[i] == target)
        {
            printf("%d", i);
            return 0;
        }
    }

    printf("Elements not found");

    return 0;
}
```

INPUT

6 7 8 1 2 3 4 5;3

OUTPUT

5

Q10. Remove Duplicates Sorted

Score: 10.00 TC: 3/3 passed

CODE

```
#include <stdio.h>

int main()
{
    int a[1000];
    int n = 0;
    int i;
    int j = 0;

    while (scanf("%d", &a[n]) == 1)
    {
        n++;
    }

    if (n > 0)
    {
        for (i = 1; i < n; i++)
        {
            if (a[i] != a[j])
            {
                j++;
                a[j] = a[i];
            }
        }

        for (i = 0; i ≤ j; i++)
        {
            if (i > 0)
                printf(" ");

            printf("%d", a[i]);
        }
    }

    return 0;
}
```

INPUT

1 2 3 4 5

OUTPUT

1 2 3 4 5